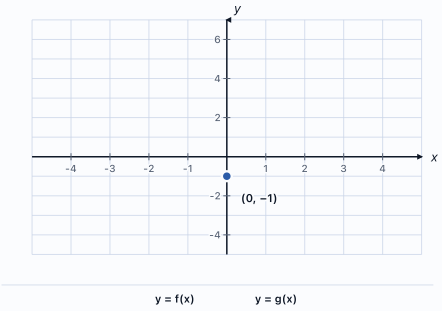
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

GAT Mental Arithmetic, Divisibility & Surds Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 6x + 20$ in the domain of GAT Arithmetic Shortcuts, Surds & Number Sense. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(5)$.



A 26
B 29
C 15
D 24

2 MCQ

A mathematical model in GAT Arithmetic Shortcuts, Surds & Number Sense is defined by the cubic function $y_2(x) = 2x^3 - 3x^2 + 15x - 8$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69
B 51
C 36
D 27

3 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, two displacement vectors are given by $\mathbf{u}_3 = \langle 3, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A −29
B 73
C 44
D 156

4 MCQ

The linear growth profile shown in the diagram for GAT Arithmetic Shortcuts, Surds & Number Sense follows an arithmetic progression with first term $a_1 = 4$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 88

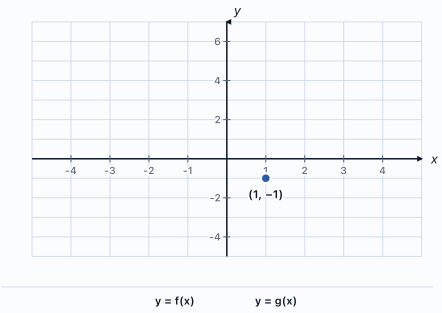
B 780

C 1320

D 690

5 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 12x)\,\mathrm{d}x$.



A 72

B 96

C 84

D 37

6 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, consider the quadratic function $f_6(x) = x^2 - 8x + 99$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(11)$.

A 219

B 215

C 132

D 208

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 4x^2 + 33x - 15$ studied in GAT Arithmetic Shortcuts, Surds & Number Sense. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 417

B 320

C 353

D 73

8 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, two displacement vectors are given by $\mathbf{u}_{8} = \langle 4, -7 \rangle$ and $\mathbf{v}_{8} = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_{8} \cdot \mathbf{v}_{8} + \|\mathbf{u}_{8}\|^{2}$.

A -3

B 65

C 132

D 62

9 MCQ

What is the units digit of 2^{17}?

A 6

B 8

C 2

D 4

10 MCQ

What is the units digit of 3^{21}?

A 3

B 9

C 1

D 7

11 MCQ

What is the units digit of 7^{23}?

A 9

B 7

C 3

D 1

12 MCQ

What is the units digit of 8^{27}?

A 2

B 4

C 8

D 6

13 MCQ

What is the units digit of 2^{29}?

A 8

B 2

C 4

D 6

14 MCQ

What is the units digit of 3^{33}?

A 1
B 7
C 9
D 3

15 MCQ

What is the units digit of 7^{35}?

A 1
B 7
C 3
D 9

16 MCQ

What is the units digit of 8^{39}?

A 6
B 2
C 4
D 8

17 MCQ

What is the units digit of 2^{41}?

A 6
B 4
C 8
D 2

18 MCQ

What is the units digit of 3^{45}?

A 1
B 7
C 3
D 9

19 MCQ

What is the units digit of 7^{47}?

A 9
B 7
C 1
D 3

20 MCQ

What is the units digit of 8^{51}?

A 6
B 4
C 2
D 8

21 MCQ

What is the units digit of 2^{53}?

A 2
B 6
C 4
D 8

22 MCQ

What is the units digit of 3^{57}?

A 9
B 3
C 1
D 7

23 MCQ

What is the units digit of 7^{59}?

A 3
B 7
C 1
D 9

24 MCQ

What is the units digit of 8^{63}?

A 2
B 8
C 6
D 4

25 MCQ

What is the units digit of 2^{65}?

A 2
B 6
C 4
D 8